F18CE – Midterm test 2

Multiple choice questions

1. (1 point) Let $\{a_r\}$ be a decreasing sequence with $a_r \geq 0$ and such that $\lim_{r\to\infty} a_r = 0$. Choose all the correct statements:

   ◯ The series $\sum_{r=1}^{\infty} a_r$ converges.

   √ **The series $\sum_{r=1}^{\infty}(-1)^{r+1}a_r$ converges.**

   ◯ The series $\sum_{r=1}^{\infty} a_r$ diverges.

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   √ **We can't say if $\sum_{r=1}^{\infty} a_r$ converges.**

   ◯ We can't say if $\sum_{r=1}^{\infty}(-1)^{r+1}a_r$ converges.

   Solution: Consider the counterexamples $a_n = \frac{1}{n}$ for a), and $a_n = \frac{1}{n^2}$ for c), which prove e). b) is a theorem in the notes.

2. (1 point) Let $a < b$ be real numbers, and let f be a continuous function defined on $I = [a, b]$. Then

   √ **f is bounded on $[a, b]$,** √ **f is Riemann integrable on $[a, b]$,** ◯ f is differentiable on (a, b).

3. (1 point) Suppose g is differentiable on (a, b). If $c \in (a, b)$ and $g'(c) = 0$, then $g(c)$ is either the maximum or the minimum of g on (a, b).

   ◯ True √ **False**

4. (1 point) If $g : [a, b] \to \mathbb{R}$ is differentiable at a point $c \in (a, b)$, then g is continuous at c.

   √ **True** ◯ False

5. (1 point) $\sum_{r=1}^{\infty} a_r$ converges if and only if $\lim_{r->\infty} a_r = 0$.

   ◯ True √ **False**

6. (1 point) The interval of convergence for the power series $\sum_{r=1}^{\infty} \frac{3^r}{r} x^r$ is

◯ $(-3,3)$ ◯ $\left(-\frac{1}{3},\frac{1}{3}\right)$ √ $\left[-\frac{1}{3},\frac{1}{3}\right)$ ◯ $\left(-\frac{1}{3},\frac{1}{3}\right]$

Solution: $\lim_{r\to\infty} \frac{3^{r+1}r}{(r+1)3^r} = \lim_{r\to\infty} 3\frac{r}{r+1} = 3$, then the radius of converges is $R = \frac{1}{3}$. At $x = -\frac{1}{3}$, the series is $\sum_{r=1}^{\infty} \frac{(-1)^r}{r}$, which converges, whereas at $x = \frac{1}{3}$ we have $\sum_{r=1}^{\infty} \frac{1}{r}$, which diverges.

7. (1 point) The power series $\sum_{r=1}^{\infty} (3-x)^r$ converges absolutely on the interval

◯ $(-3,3)$ √ $(2,4)$ ◯ $[2,4]$

Solution: The series $\sum_{r=1}^{\infty} x^r$ converges for $x \in (-1,1)$. Therefore, the series $\sum_{r=1}^{\infty}(3-x)^r$ converges if $3 - x \in (-1,1)$, i.e. $x \in (2,4)$. Observe that the interval cannot be closed because the series for $x = 2$ and $x = 4$ are $\sum_{r=1}^{\infty} 1$ and $\sum_{r=1}^{\infty}(-1)^r$ respectively.

8. (1 point) If P and Q are partitions of $[a,b]$, then $P \cup Q$ is a refinement of both P and Q.

√ **True** ◯ False

9. (1 point) If $f : [a,b] \to \mathbb{R}$ and P be a partition of $[a,b]$. Then f is Riemann integrable if and only if its upper and lower sums with with respect to P are equal.

◯ True √ **False**

Solution: A function is Riemmann integrable if and only if the sup of the lower sums is equal to the inf of the upper sums. For instance, take the smallest partition $P = \{a,b\}$, then $L(P,f) = U(P,f)$ if and only if the function f is constant.

10. (1 point) Consider the function f defined as

$$f(x) = \begin{cases} 1, & 0 < x \leq 1, \\ 2, & 1 < x \leq 2. \end{cases}$$

Define $G(x) = \int_0^x g(t)\,dt$, then

√ G **is continuous on** $[0,2]$**.**

◯ G is discontinuous on $[0,2]$.

◯ G is differentiable on $(0,2)$ and $G' = g$.

◯ G is differentiable on $(0,2)$ but $G' \neq g$.

√ G **is not differentiable on** $(0,2)$**.**

Solution: By applying the Fundamental Theorem of Calculus on the intervals $[0,1]$ and $[1,2]$, we can find G:

$$G(x) = \begin{cases} \displaystyle\int_0^x 1\,dt = x, & 0 < x < 1, \\ \displaystyle 1 + \int_1^x 2\,dt = 2x - 1, & 1 \le x < 2. \end{cases}$$

Looking at the side limits at $x = 1$, we see that G is continuous. However, G is not differentiable at $x = 1$ because $\lim_{h\to 0} \dfrac{G(1+h) - G(1)}{h}$ does not exist:

$$\lim_{h\to 0^-} \frac{G(1+h) - G(1)}{h} = \lim_{h\to 0^-} \frac{1+h-1}{h} = 1,$$
$$\lim_{h\to 0^+} \frac{G(1+h) - G(1)}{h} = \lim_{h\to 0^-} \frac{2+2h-1-1}{h} = 2.$$

Written problem

Let f and g be functions that are continuous on $[a,b]$ and differentiable on (a,b). Show that there exists at least one point $c \in (a,b)$ such that:

$$[f(b) - f(a)]\, g'(c) = [g(b) - g(a)]\, f'(c).$$

State explicitly every theorem you use in your proof, and why it is applicable.

Hint: define a new function $h(x) = \alpha f(x) - \beta g(x)$ on $[a,b]$ for suitable constants α and β.

Solution: Define the function $h(x) = \alpha f - \beta g$ on $[a,b]$, for some real numbers α and β. Then h is continuous on $[a,b]$ because it is the linear combination of continuous functions on $[a,b]$. Similarly, h is differentiable on (a,b) because it is the linear combination of differentiable functions on (a,b). Therefore, we can apply the Mean Value Theorem.

Mean Value Theorem. Let f be continuous on $[a,b]$ and differentiable on (a,b). Then, there exists a $c \in (a,b)$ such that $f(b) - f(a) = f'(c)(b-a)$.

Therefore, there exists a point $c \in (a,b)$ such that $h(b) - h(a) = h'(c)(b-a)$. We calculate the LHS and the RHS. Starting with $h(b) - h(a)$:

$$h(b) - h(a) = \alpha f(b) - \beta g(b) - \alpha f(a) + \beta g(a) = \alpha(f(b) - f(a)) - \beta(g(b) - g(a)).$$

On the other hand, $h'(c) = \alpha f'(c) - \beta g'(c)$. Therefore, we choose

$$\alpha = g(b) - g(a), \quad \beta = f(b) - f(a),$$

in order to have the quantity that we need on the RHS. Moreover, we have that $h(b) - h(a) = 0$, therefore

$$0 = h'(c) = [g(b) - g(a)]f'(c) - [f(b) - f(a)]g'(c),$$

which is the equality we needed.

Alternatively, you could choose α and β in such a way that gives $h(b) = h(a)$. This would give the same choice for α and β. Then, you can apply (and state) Rolle's Theorem, which guarantees the existence of a point $c \in (a,b)$ such that $h'(c) = 0$.